

[Aim: 100/100^o in Maths]

आरंभ

TRIGONOMETRY

Lecture #3

ATK²:

sin	cos	tan θ
P	B	P
H	H	B
cosec	sec	cot θ

$$\sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

Reciprocal
identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

#LP:

In $\triangle ABC$, $\angle B = 90^\circ$. If $\frac{AB}{AC} = \frac{1}{2}$, then $\cos C$ is equal to

(A) $\frac{3}{2}$

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{\sqrt{3}}$

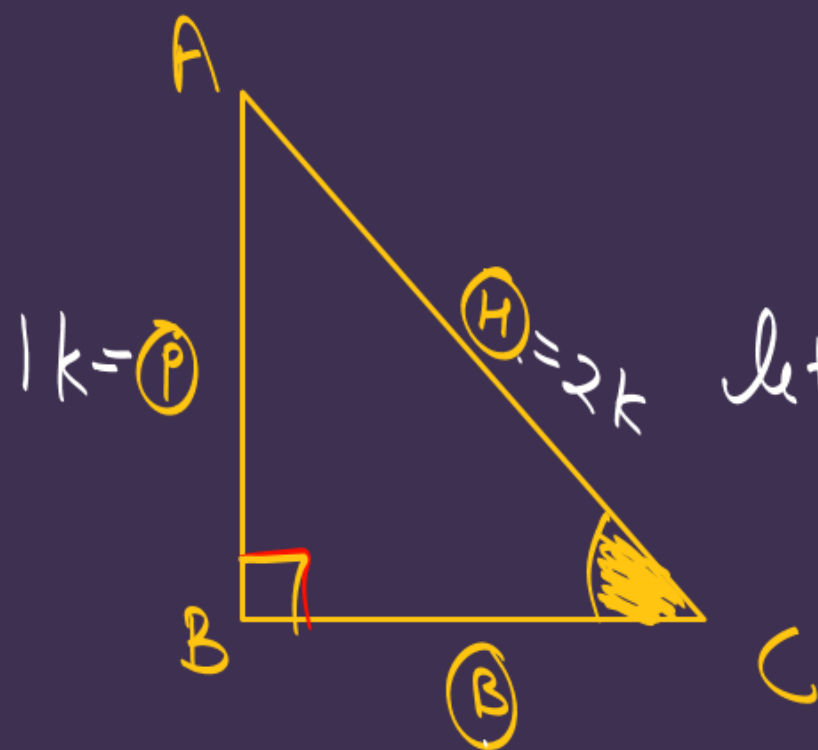
[CBSE 2025]

To find

$$\cos C = \frac{B}{H} = \frac{BC}{AC}$$

$$= \frac{\sqrt{3}K}{2K}$$

$$\cos C = \frac{\sqrt{3}}{2}$$



$$\frac{AB}{AC} = \frac{1}{2}$$

$$\text{Let } \begin{cases} AB = 1k \\ AC = 2k \end{cases}$$

Pyth $H^2 = P^2 + B^2$

$$(2k)^2 = (1k)^2 + B^2$$

$$4k^2 = k^2 + B^2$$

$$4k^2 - k^2 = B^2$$

$$B^2 = 3k^2$$

$$B = \sqrt{3k^2} \Rightarrow B = \sqrt{3}K$$

#LP: If $5 \tan \alpha = 4$, show that $\frac{5 \sin \alpha - 3 \cos \alpha}{5 \sin \alpha + 2 \cos \alpha} = \frac{1}{6}$

$$\tan \alpha = \frac{4}{5}$$

(MII)

LHS

div Num & Den by $\cos \alpha$

$$\Rightarrow \frac{5 \frac{\sin \alpha}{\cos \alpha} - 3 \frac{\cancel{\cos \alpha}}{\cancel{\cos \alpha}}}{5 \frac{\sin \alpha}{\cos \alpha} + 2 \frac{\cancel{\cos \alpha}}{\cancel{\cos \alpha}}}$$

$$\Rightarrow \frac{5 \tan \alpha - 3}{5 \tan \alpha + 2}$$

$$\Rightarrow \frac{5 \times \frac{4}{5} - 3}{5 \times \frac{4}{5} + 2} \Rightarrow \frac{1}{6} = \text{RHS} \quad \text{H.P.}$$

(MI)

$$\tan \alpha = \frac{P}{B} = \frac{4}{5}$$

$$P = 4k$$

$$B = 5k$$

H = Pytho

given

$\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$

1st P:- If $\cot \beta = \frac{2}{3}$, then find the value of $\frac{3 \sin \beta + 2 \cos \beta}{3 \sin \beta - \cos \beta}$

$$\cot \beta = \frac{2}{3} \quad \checkmark$$

↓

$$\frac{\cos \beta}{\sin \beta}$$

div Num & Den by $\sin \beta$

$$\Rightarrow \frac{\frac{3 \cancel{\sin \beta}}{\cancel{\sin \beta}} + 2 \left(\frac{\cos \beta}{\sin \beta} \right)}{\frac{3 \cancel{\sin \beta}}{\cancel{\sin \beta}} - \left(\frac{\cos \beta}{\sin \beta} \right)}$$

$$\Rightarrow \frac{3 + 2 \cot \beta}{3 - \cot \beta} \rightarrow$$

$$\frac{3 + 2 \left(\frac{2}{3} \right)}{3 - \left(\frac{2}{3} \right)} \rightarrow \frac{\frac{9+4}{3}}{\frac{9-2}{3}} \Rightarrow \frac{13}{7}$$

#LP:

#Trigonometric Ratios of some Specific Angles:

$\theta^\circ \rightarrow$	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\frac{\sin \theta}{\cos \theta} \leftarrow \tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D.
$\frac{1}{\sin \theta} \leftarrow \operatorname{cosec} \theta$	N.D.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\frac{1}{\cos \theta} \leftarrow \sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D.
$\frac{1}{\tan \theta} \leftarrow \cot \theta$	N.D.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

#LP: Evaluate the following expressions:

$$\sin^2 \theta = (\sin \theta)^2$$

$$2\sin^2 30^\circ \tan 60^\circ - 3\cos^2 60^\circ \sec^2 30^\circ$$

$$\Rightarrow 2(\sin 30)^\circ (\tan 60) - 3(\cos 60)^\circ (\sec 30)^\circ$$

$$\Rightarrow 2\left(\frac{1}{2}\right)^2 (\sqrt{3}) - 3\left(\frac{1}{2}\right)^2 \left(\frac{2}{\sqrt{3}}\right)^2$$

$$\Rightarrow \left(\frac{1}{2}\right)^2 \left[2 \times \sqrt{3} - 3 \times \frac{4}{3} \right]$$

$$\Rightarrow \frac{1}{4} [2\sqrt{3} - 4]$$

$$\Rightarrow \frac{1}{2} \times 2 (\sqrt{3} - 2) \Rightarrow$$

$$\frac{1}{2} (\sqrt{3} - 2) \sim$$

#LP:

Evaluate the following:

$$\frac{3 \sin 30^\circ - 4 \sin^3 30^\circ}{2 \sin^2 50^\circ + 2 \cos^2 50^\circ}$$

[CBSE 2025]

$$\overset{00}{(ii)} \quad \frac{3 \cos^2 60 - 2 \tan 30}{1 + \sin^2 45}$$

$$\Rightarrow \frac{3 \left(\frac{1}{2}\right)^2 - 2 \left(\frac{1}{\sqrt{3}}\right)}{1 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

↓

$$\overset{00}{(iii)} \quad \frac{2 \tan 45 + \cot 45}{\sin^2 90}$$

$$\Rightarrow \frac{2(1) + (1)}{(1)^2}$$

$$\Rightarrow \underline{\underline{3}}$$

#LP:

If $x \tan 60^\circ \cos 60^\circ = \sin 60^\circ \cot 60^\circ$, then $x =$

[CBSE SQP (Standard) 2022-23]

- (A) $\cos 30^\circ = \frac{\sqrt{3}}{2}$ ✗ ☒ (B) $\tan 30^\circ = \frac{1}{\sqrt{3}}$
(C) $\sin 30^\circ = \frac{1}{2}$ ✗ (D) $\cot 30^\circ$

$$x \tan 60 \cos 60 = \sin 60 \cot 60$$

$$x (\sqrt{3}) \left(\frac{1}{2}\right) = \left(\frac{\sqrt{3}}{2}\right) \left(\frac{1}{\sqrt{3}}\right)$$

$$x \sqrt{3} = 1$$

$$x = \frac{1}{\sqrt{3}}$$

K3B

If $\sin A = \sin B$
then $\angle A = \angle B$

slly
Same T-ratio

$\cos A = \cos B$
 $\angle A = \angle B$

$\tan A = \tan B$
 $\angle A = \angle B$

⋮

~~$\sin A = \tan B$
 $\angle A \neq \angle B$~~

converse is also true
if $\angle A = \angle B$

taking sin both sides
 $\sin A = \sin B$

cos both side
 $\cos A = \cos B$
 $\tan A = \tan B$
 $\operatorname{cosec} A = \operatorname{cosec} B$

⋮

QP:- If $\sin \alpha = \frac{\sqrt{3}}{2}$ find the measure of α ??

$\sin \alpha = \sin 60^\circ$
on comparing,

$$\therefore \boxed{\alpha = 60^\circ}$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$

$$\cos A = \frac{1}{2}, \angle A = ?$$

$\cos A = \cos 60^\circ$
on comparing,

$$\boxed{A = 60^\circ}$$

$$\cot(2\beta) = \sqrt{3}, \angle \beta = ?$$

$$\cot(2\beta) = \cot 30^\circ$$

on comparing,

$$2\beta = 30^\circ$$

$$\boxed{\beta = 15^\circ}$$

15° ✓

#LP: Find the value of θ in each of the following:

(i) $2 \sin 2\theta = \sqrt{3}$

$$\sin 2\theta = \left(\frac{\sqrt{3}}{2} \right)$$

$$\sin 2\theta = \sin 60^\circ$$

on compary

$$2\theta = 60^\circ$$

$$\theta = 30^\circ$$

(ii) $\sqrt{3} \tan 2\theta - 3 = 0$

$$\sqrt{3} \tan 2\theta = 3$$

$$\tan 2\theta = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\tan 2\theta = \frac{3\sqrt{3}}{\sqrt{3}}$$

$$\tan 2\theta = \sqrt{3}$$

$$\tan 2\theta = \tan 60^\circ$$

on compary

$$2\theta = 60^\circ$$

$$\theta = 30^\circ$$

#LP: Find the value of x in each of the following:

(i) $\tan 3x = \sin 45^\circ \cos 45^\circ + \sin 30^\circ$

$$\tan 3x = \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)$$

$$\tan 3x = \frac{1}{2} + \frac{1}{2}$$

$$\tan 3x = 1$$

$$\tan 3x = \tan 45^\circ$$

on compare

$$3x = 45^\circ \Rightarrow x = 15^\circ$$

#LP:

If $\sqrt{2} \sin(60^\circ - \alpha) = 1$, then α is [CBSE 2011]

(A) 45°

☒ (B) 15°

(C) 60°

(D) 30°

$$\sin(60 - \alpha) = \frac{1}{\sqrt{2}}$$

$$\sin(60 - \alpha) = \sin 45$$

on comparing

$$60 - \alpha = 45$$

$$60 - 45 = \alpha$$

$$\boxed{\alpha = 15^\circ} \quad \text{H}$$

#LP: If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, $0^\circ < A + B \leq 90^\circ$ and $A > B$, then find the measures of angles A and B.

$$\sin(A + B) = 1$$

$$\sin(A + B) = \sin 90$$

on compary

$$A + B = 90^\circ \quad \text{--- (I)}$$

$$\cos(A - B) = \frac{\sqrt{3}}{2}$$

$$\cos(A - B) = \cos 30$$

on compary.

$$A - B = 30^\circ \quad \text{--- (II)}$$

$$\begin{aligned} A &= 60^\circ \\ B &= 30^\circ \end{aligned} \quad \text{A}$$

HW #LP: In an acute angled triangle ABC, if $\tan (A+B-C)=1$ and $\sec(B+C-A)=2$, find the value of A,B and C.

THANK YOU COODIESS !!

